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Habitat choice of Red-breasted Flycatchers *Ficedula parva* is dependent on forestry management and game activity in a deciduous forest in Vienna (Austria)

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Capsule Forest management and game populations have a major impact on the distribution of Red-breasted Flycatchers.

Aims To improve the understanding of habitat choice by Red-breasted Flycatchers and to provide a basis for nature conservation measures.

Methods The abundance and distribution of Red-breasted Flycatchers were investigated on 23 survey areas of approximately 150 ha. Habitat structure was measured at 42 points in territories and at 151 control points outside territories. Differences in forest structure between territories and control points were highlighted using discriminant function analyses.

Results Territories of Red-breasted Flycatchers correlated positively with the portion of hornbeam, the age of the stands and volume of lying dead trunks. Territories were mainly found in stands with high levels of mammalian game activity and accordingly a more sparse bush layer and less ground cover. Territories were closer to ditches than to control points. Red-breasted Flycatchers avoided stands under intensive forestry management.

Conclusion Along with forestry management, the activity of mammalian game has major impacts on forest structure. Accordingly, the effect of mammalian game activity can have important implications for nature conservation.

Forests in central Europe are mainly influenced by forestry use, but the effect of game populations on forest structure is gaining importance, and both influence nature conservation (Scherzinger 1996). Game species can cause changes in forest communities by browsing and by indirect energy through-flow (McShea & Rappole 2000). Bird species are particularly sensitive to changes in vegetation amount and composition (Mills *et al.* 1991). Accordingly, in a variety of instances nature conservation measures can be successful only if the interactions between forestry use, game activity and bird species are well understood. Here we provide data of interactions between forestry use, game activity and Red-breasted Flycatcher *Ficedula parva* habitat use.

In Austria the Red-breasted Flycatcher is at the western end of its distribution (Glutz von Blotzheim & Bauer 1993) and its breeding population is estimated at 1500–3000 pairs (BirdLife International 2004). About 8–10% of the national population is found in Vienna (Wichmann & Frank 2003). In Austria Red-breasted Flycatchers are mostly found in the northern parts of the Alps in Beech *Fagus sylvatica* and Hornbeam *Carpinus betulus* forests (Dvorak *et al.* 1993); however, they can occupy a wide range of forest habitats, ranging from almost purely deciduous to almost purely coniferous stands (Glutz von Blotzheim & Bauer 1993). Our aim was to frame general principles for habitat selection by Red-breasted Flycatchers and study the effects of forestry use and game density on the habitat. The results have implications for nature conservation activities that support populations of Red-breasted Flycatchers.

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METHODS

Study area

The study was conducted in the Viennese Woods, which lie within the borders of the city of Vienna and cover an area of 54 km². They are dominated by deciduous trees, mainly oaks *Quercus* sp., beech and hornbeam. Because of its proximity to the city, the forest is a popular recreation area; forest management reflects this and is not intensive in terms of wood production. The wooded areas can be separated into two major regions. The Lainzer Tiergarten covers an area of 22 km² that is protected under NATURA 2000, and is composed mostly of old stands (up to 300 years), especially oaks. The remaining area has been used for more intensive forestry in the last few decades and trees are mostly younger.

Study design

We mapped the locations of singing males, because the habitat used by singing males was linked to the availability of breeding sites and the occurrence of successful nests (Tjernberg 1984). Red-breasted Flycatchers were recorded at 23 randomly selected survey areas, during two visits between the end of April and mid-June. The areas were about 150 ha in size, but this varied because we did not survey within settlements or agricultural areas. A total of 2997 ha were surveyed.

Structural measurements of the stands were made to describe the habitat of the Red-breasted Flycatchers. The parameters chosen are either described in literature or are suspected of being important predictors of the occurrence of the species. Structure was measured at points where singing males were noted, and at so-called 'control points' outside the nominal territories. Control points were randomly selected from throughout the survey areas and were at least 100 m away from the song-posts where flycatchers were recorded, because Red-breasted Flycatchers usually stay within 30–50 m of the potential nest-site (Steinfatt 1937, Tjernberg 1984). A total of 42 territories and 151 control points were included in the analysis.

To determine tree density, the point-centred quarter (PCQ) method was used (Kreeb 1983):

$$D = \frac{100}{E^2}; \text{ where } E =$$

$$\frac{\sum (\text{distance from point to closest tree in each quadrant})}{4}$$

The mean tree diameter at breast height (dbh) and mean height to the base of the crown were calculated for the four trees closest to the randomly selected control point. The base of the crown was defined as the leaf-bearing branch closest to the ground. The number of dead branches below and above the base of the crown was recorded in each of the trees. If any of the four trees measured by us had a diameter of 10 to 20 cm, the point was defined as rejuvenating. As there is evidence that territories of Red-breasted Flycatchers are located on wet and moist forest land close to water (Glutz von Blotzheim & Bauer 1993), the distance to the nearest ditch from the random point was also measured.

Following the methods of the local forest administration, the vertical section of the forest, the quantity of dead trunks and the influence of mammalian game on the forest were measured in an 8-m-radius circle around each point (Eckmüller *et al.* 2001). To get an idea of foliage profile in different strata, the vertical section of the forest vegetation was divided into bush layer (0–1.3 m), low level layer (1.3 m to one-third the height of the tree stand), mid-level layer (one-third to two-thirds the height of the tree stand) and crown closure layer (two-thirds to the top of stand height). We measured the crown closure and density of the bush layer by looking at these layers through an eye tube (made out of a toilet-roll tube) every five steps taken in a circle around the survey point (Cyr & Oelke 1976, Bibby *et al.* 1992). If leaves appeared in the field of vision the crown was defined as 'closed', if not it was defined as 'open'. This procedure was performed ten times and the proportion of crown observations that were classified as closed was calculated. The same procedure, but looking downwards, was used to evaluate the proportion of ground churned up by game (mainly Wild Boar *Sus scrofa*). In the following this variable will be termed 'game activity'. The number and volume of standing dead trees and the number and volume of lying dead timber that had a diameter >20 cm at mid-length were also determined.

Tree species composition, the age of the stand and measures of the intensity of forestry were provided by the forest administration. The intensity of forestry within forest units was measured by the portion of the area managed; units were 0.37–31.7 ha in size. We compared the forestry units surrounding control points with those around territories. Solar radiation at each point was derived from values from the Central Institute of Meteorology and Geodynamics, Vienna, adjusted for exposition and slope (Bruck *et al.* 1983)

Altitude was not considered because the differences in altitudes were negligible.

Statistical analysis

A discriminant function analysis was used to test for differences between control points and Red-breasted Flycatcher territories. The discriminant function was computed by using logistic regressions (Sokal & Rohlf 1995). The significance of the coefficients was tested by randomization, as were correlations between each variable and the principal component axis. Two thousand resamplings were performed.

The variables used in the multivariate analysis and the manner in which they were transformed (Sokal & Rohlf 1995) are listed in Table 1. Variables showing high correlations were factorized for the purpose of avoiding colinearity. High correlations were found between bush layer, density of ground cover and game activity; thus one factor was formed out of these variables (Table 1). Game activity is negatively correlated with component 1, whereas bush layer and the density of ground cover show a positive relationship. This means that stands with high game activity have less bush-layer vegetation and ground cover. This factor is referred to as 'effect of game activity'. A similar procedure was performed with the canopy layers above 1.3 m to aggregate the different layers (Table 2). While component 1 correlated positively with low-level and mid-level vegetation layers, component 2 showed a high correlation with the crown closure layer. Component 1 is referred to as 'mid-canopy' and component 2 is referred to as 'top canopy'.

Logistic regressions were carried out using the program LOGIREG (H. Nemeschkal, University of Vienna). Data on forest management were not available for the whole area, thus these data were analysed with bivariate methods. Bivariate analyses were performed in Simstat 2.5. $P < 0.05$ was considered to be significant.

RESULTS

Population and density

Overall, 109 territories were located in 23 survey areas, suggesting an average population density of 3.6 territories/km² (Fig. 1). About 80% of territories were found in the Lainzer Tiergarten, where the average density was 6.9 territories/km². The densities of the survey areas ranged between 0.0 and 10.3 territories/km².

Table 1. Factorization of percentage bush layer, ground cover and game activity. The correlation matrix for the variables and the percentage of explanation of the form factor are presented.

	Component 1, 'effect of game activity'
Bush layer	0.776
Ground cover	0.850
Game activity	-0.833
Per cent explained	67.26

Table 2. Factorization of percentage canopy layers. The correlation matrix for the variables and the percentage of explanation of the form factor are presented.

	Component 1, 'mid-canopy'	Component 2, 'top canopy'
Low-level layer	0.838	-0.186
Mid-level layer	0.817	0.281
Crown closure layer	-0.077	0.966
Per cent explained	45.83	34.92
Cumulative per cent explained	45.83	80.75

Habitat structure

Five variables were found to contribute significantly to the overall model: distance to the nearest ditch, proportion of hornbeam, volume of lying dead trunks, 'effect of game activity' factor, and the age of stands (Table 3). Territories were significantly closer (median 30 m) to ditches than to control points (median 80 m, $\beta = -1.690$, $P < 0.0005$, Table 4). In all, 47.6% of the territories were within 20 m of the nearest ditch. Red-breasted Flycatchers preferred stands with a high proportion of hornbeam ($\beta = 0.061$, $P < 0.0005$). In Red-breasted Flycatcher territories the median amount of Hornbeam was 30.5%; at control points the median was 6%. Additionally, stands occupied by Red-breasted Flycatchers were significantly older (median 136 years) than stands containing control points (median 94 years) ($\beta = 2.120$, $P = 0.022$, Fig. 2). Stands younger than 44 years of age were not inhabited. Territories were negatively correlated with the factor 'effect of game activity' ($\beta = -2.030$, $P < 0.0005$), meaning that they had higher activity of mammalian game and lower density of bush layer and ground cover (Table 4). Activity of game species was found in 86% of all territories, but only in 36% of the control points. In 88% of the territories the bush layer was absent; at control points this value was 44%. In 88% of the territories ground cover was missing, in contrast to 31% of the control points. Lying dead trunks were found in

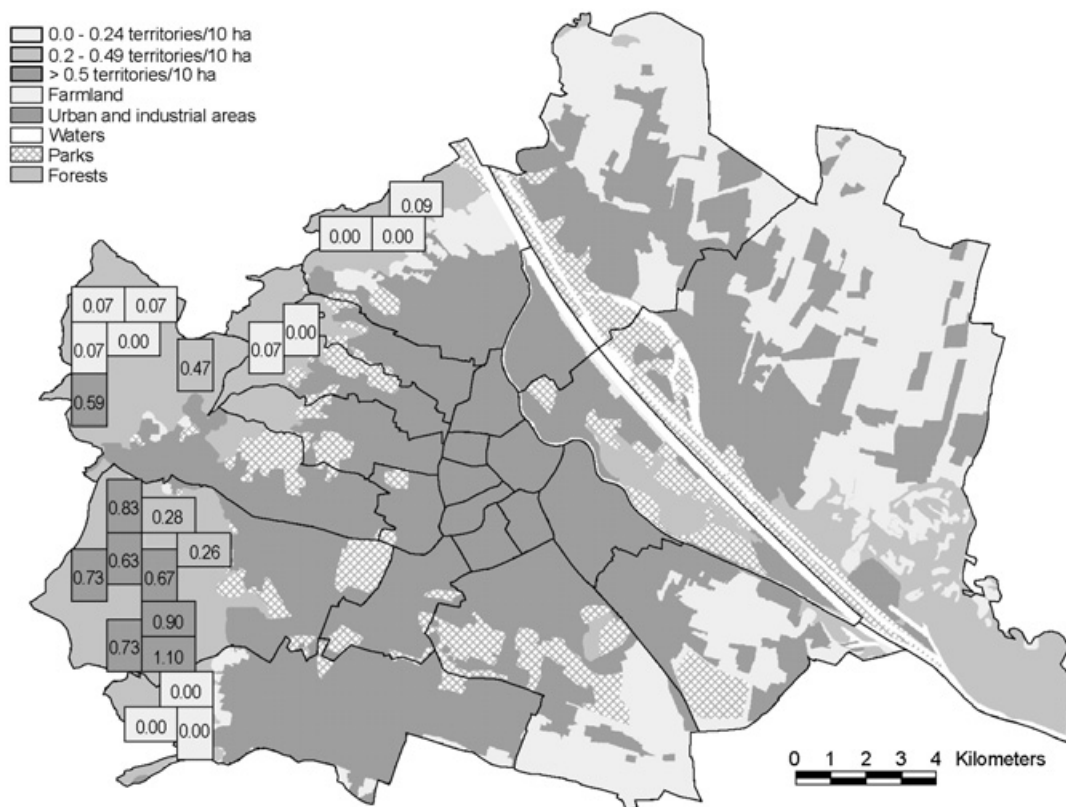


Figure 1. Distribution and density (territories/10 ha) of Red-breasted Flycatchers in the forests of Vienna. Use of map kindly authorized by Magistratsabteilung 22 – Umweltschutz, Vienna.

Table 3. Results of the discriminant function analysis of the variables with means of logistic regression.

	Beta	P
Intercept	-10.563	0.055
Distance to nearest ditch (m)	-1.690	<0.0005
Mean dbh (no./100 m ²)	-1.054	0.859
Tree diameter (cm)	3.224	0.111
No. standing dead trunks	-0.088	0.761
Lying dead trunks (m³)	4.184	0.012
No. dead branches	-0.102	0.472
Oaks (%)	-0.004	0.697
Beeches (%)	0.012	0.293
Hornbeams (%)	0.061	<0.0005
Age of the stand (years)	2.120	0.022
Irradiation (W/m ²)	-0.008	0.657
Lower edge of the crown (m)	1.636	0.081
Factor 'effect of game activity'	-2.025	<0.0005
Mid-canopy (%)	-0.229	0.131
Top canopy (%)	0.319	0.148

Significance was tested by randomization. One-tailed significance for deviance (saturated-observed model), $P < 0.0005$. One-tailed significance for log-likelihood function (observed model), $P < 0.0005$. Significant variables are in bold (two-tailed test). Territories, $n = 42$, control points, $n = 151$.

69% of the territories but only in 34% of the control points.

Red-breasted Flycatchers avoided stands where forest management took place. Forestry units surrounding control points were managed in 50.7% of cases. In contrast, only 36% of the units around territories were used by forestry. At the control points the median proportion of area used by forestry was 50%, while in territories it was only 25%, and this difference was significant ($U = 2201.5$, $n_1 = 135$, $n_2 = 42$, $P = 0.029$). Red-breasted Flycatchers avoided stands where thinning occurred ($U = 2149.0$, $n_1 = 135$, $n_2 = 42$, $P = 0.008$).

DISCUSSION

The occurrence of Red-breasted Flycatchers in Vienna is not restricted to any single forest type. With the exception of pure oak forests, the species was found in a variety of forests ranging from pure beech to oak/hornbeam. This is consistent with published information that Red-breasted Flycatchers are able to

Table 4. Descriptive statistics of the variables used in the logistic regression.

	Trans	Control points (n = 151)			Territories (n = 42)		
		Mean $\pm$ sd	Median	Range	Mean $\pm$ sd	Median	Range
Distance to the nearest ditch (m)	log	100.4 $\pm$ 87.2	80	1–350	62.1 $\pm$ 74.7	30	1–350
Tree density (no./100 m ²)	no	6.32 $\pm$ 5.29	4.88	0.2–30.9	5.5 $\pm$ 4.2	4.16	0.8–16.3
Mean dbh (cm)	log	39.82 $\pm$ 12.46	37.5	11.5–90.5	47.5 $\pm$ 16.8	46.4	20–81.3
No. standing dead trunks	sqr	2.7 $\pm$ 4.0	1	0–20	2.5 $\pm$ 2.9	1.5	0–12
Lying dead trunks (m ³)	log	0.2 $\pm$ 0.5	0	0–3.66	0.3 $\pm$ 0.4	0.16	0–1.52
No. dead branches	sqr	23.7 $\pm$ 18.7	18	2–89	31.6 $\pm$ 21.5	26.5	3–75
Oaks (%)	arcsin	29.3 $\pm$ 30.5	20	0–100	34.5 $\pm$ 23.3	41.5	0–100
Beeches (%)	arcsin	39.8 $\pm$ 36.2	30	0–100	31.6 $\pm$ 31.1	23.5	0–100
Hornbeams (%)	arcsin	12.9 $\pm$ 16.7	6	0–70	28.6 $\pm$ 23.5	30.5	0–82
Age of the stand (years)	log	102.9 $\pm$ 67.7	94	7–382	196.4 $\pm$ 1180	136	44–382
Irradiation (W/m ²)	no	133.3 $\pm$ 12.4	139.1	88.5–15	129.3 $\pm$ 9.4	131.7	103.4–145.4
Lower edge of the crown (m)	log	4.3 $\pm$ 3.0	4	0–12	4.2 $\pm$ 2.9	3	1.5–14.0
Mid-canopy (%)	arcsin	63.4 $\pm$ 24.5	60	10–100	67.1 $\pm$ 22.9	70	10–100
Top canopy (%)	arcsin	77.0 $\pm$ 21.1	80	10–100	79.3 $\pm$ 16.4	80	40–100
Factor 'effect of game activity'							
Churned ground (%)	arcsin	12.6 $\pm$ 23.1	0	0–100	55.2 $\pm$ 31.6	60	0–100
Ground cover (%)	arcsin	26.5 $\pm$ 29.5	20	0–100	5.48 $\pm$ 17.3	0	0–90
Bush layer (%)	arcsin	20.6 $\pm$ 27.4	10	0–100	2.9 $\pm$ 11.3	0	0–70

Trans, transformation; log, logarithmic; sqr, square root; no, no transformation.

populate a wide range of forest types, including beech forests, oak/hornbeam forests, spruce forests, and even orchards (Steinfatt 1937, Weber 1958, Tjernberg 1984, Fuxa 1991, Glutz von Blotzheim & Bauer 1993, Feldner & Rass 1999). Nevertheless, in the Viennese Woods Red-breasted Flycatchers showed an obvious preference for hornbeam (Tables 1 & 2).

Similar relationships have been found by other authors (Steinfatt 1937, Sturm 1986, Fuxa 1991, Cezary & Soćko 2004). In Białowieca National Park in eastern Poland, hornbeams were mostly used as breeding places, as 47% of the nests were built in hornbeams (Cezary & Soćko 2004). Similar results were found in the Lainzer Tiergarten, where a majority of

nests were built in hornbeams (Fuxa 1991). Hornbeams have a large number of cavities and crotches created by side shoots, which provide many possible breeding places for Red-breasted Flycatchers. Besides providing good nesting places, hornbeams have a higher leaf area index (LAI) than other species such as Birch *Betula pendula*, pine *Pinus* sp. or oak (Hagemeier 2002). The hornbeam's high LAI is due to its high number of side shoots and the tolerance of the leaves to shade. For Red-breasted Flycatchers a high LAI is important because they glean much of their food from leaves (Weber 1958, Fuxa 1991). However, a high LAI need not be directly correlated with high crown closure as Red-breasted Flycatchers were found in open oak/hornbeam forests as well. Accordingly, the vertical vegetation structure of mid- and top canopy did not contribute significantly to the model.

In agreement with our results, Fuxa (1991) described Red-breasted Flycatchers as regularly breeding in open oak/hornbeam forests of the Lainzer Tiergarten and as hunting regularly in the crowns of old hornbeams. This seems to be inconsistent with the dense habitats (due to high crown closure) described by most other authors (Steinfatt 1937, Weber 1958, Tjernberg 1984, Glutz von Blotzheim & Bauer 1993). However, in eastern parts of the breeding range Red-breasted Flycatchers are found in relatively open habitats, including oak/hornbeam forests and even orchards (Somow 1897, Gania & Litwak 1961, both cited in Glutz von

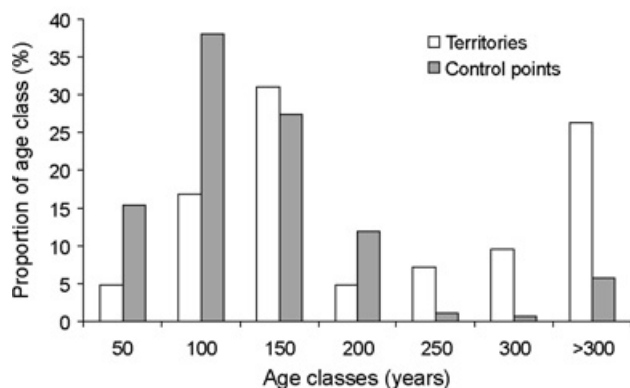


Figure 2. Frequency distribution of age classes of stands containing territories (n = 42) and control points (n = 151).

Blotzheim & Bauer 1993). Distilling our findings with the published information, we conclude that a large leaf surface is more important than dense crown closure to Red-breasted Flycatchers.

The age of the stands has an important, positive influence on the population of Red-breasted Flycatchers. The median of 136 years corresponds to the preferred age of at least 100 years mentioned by several authors (Tjernberg 1984, Fuxa 1991, Glutz von Blotzheim & Bauer 1993, Németh 2000), and 43% of the territories were situated in forests exceeding 200 years. Thus any trend towards shortening the cutting cycle would have a major negative influence on Red-breasted Flycatchers. From an ecological point of view, forest age is related to a number of detailed characteristics of the forest itself. Walter (1971) states that a 50-year-old forest has a living plant biomass of 200 t/ha, but that the biomass of a 200-year-old stand exceeds 400 t/ha. This means that beside the stem mass, the available leaf surface increases with forest age as well. A higher leaf surface improves the habitat quality for Red-breasted Flycatchers, as mentioned above. Furthermore, the amount of dead wood, the number of epiphytes and the roughness of the bark increase with age, resulting in stands that are more attractive to insects, spiders and mites (Harding & Rose 1986, Remmert 1991), all of which are potential prey for Red-breasted Flycatchers. Dead wood is especially important for a variety of invertebrates, some of which spend their whole life cycle there (Scherzinger 1996). Nearly 20% of the forest's fauna need dead wood, or feed on fungi or insects that need it (Weiss 1984, Harding & Rose 1986). Accordingly, Red-breasted Flycatchers prefer stands with a high level of lying dead trunks.

The larger amounts of rotting segments of trees, broken trunks and cracks in the bark in older stands also increase the availability of breeding places for Red-breasted Flycatchers (Wesołowski 1989, Walankiewicz 1991). Tjernberg (1984) assumes that stands younger than 50 years are avoided due to the scarcity of suitable nesting places. Stands with little or no management have a high number of potential nesting places (Sandström 1992). In line with this we found a negative relationship between intensity of forestry management and the occurrence of Red-breasted Flycatchers, which tended to avoid stands where forestry management took place. In particular, forest thinning proved to have major negative effects on the species, in agreement with the findings of Sachslehner (1992) and Tjernberg (1984).

In 90% of the territories the bush layer was absent. Besides collecting prey from the surface of leaves, Red-breasted Flycatchers hunt below the lower edge of the crown and they use the forest floor as well (Fuxa 1991, Mieslinger 1992). The literature is not entirely clear on the importance of the area under the lower edge of the crown. Weber (1958) reports that 15% of hunting takes place at the forest floor and 20% in flight. During bad weather conditions Red-breasted Flycatchers spend most of their hunting time below the crown. In contrast, however, Tjernberg (1984) reports that the density and species composition of the bush and field layer are probably not important to the selection of habitat by Red-breasted Flycatchers.

The inhibition of the development of a bush layer is mostly influenced by game activity (Belsky & Blumenthal 1997). Especially in the Lainzer Tiergarten, where nearly 80% of the population of Red-breasted Flycatchers is situated, there is a high density of game including about 800 Wild Boar, 200 European Fallow Deer *Dama dama*, 200 Mouflons *Ovis ammon musimon*, 70 Red Deer *Cervus elaphus* and 80 Roe Deer *Capreolus capreolus* in an area (fenced) of 2.440 ha (A. Mrkvicka, pers. comm.). This leads to the absence of a bush layer across large areas. Additionally, the activity by Wild Boar of breaking stems promotes the production of side shoots by hornbeams, which leads to an increase of crown volume and leaf surface (A. Mrkvicka & U. Nopp pers. comm.). However, it should be taken into consideration that due to changes in terrestrial food webs and the structure of the stands, several animal and plant species may be negatively influenced (Scherzinger 1996, McShea & Rappole 2000). Thus a high density of game can have a negative influence on invertebrates (potential prey) living on the forest floor. Knowledge about the influence of game on the ecological dynamics of forest is fragmentary (Scherzinger 1996).

Nearly half of the territories of Red-breasted Flycatchers were found within 20 m of the nearest ditch. Glutz von Blotzheim & Bauer (1993) and Tjernberg (1984) stress the importance of moist and fresh areas for Red-breasted Flycatchers; however, the birds' preference for such areas may be because stands close to ditches are usually less intensively used by forestry (Wichmann & Frank 2003).

In conclusion, forest management and the influence of game have major impacts on the occurrence of Red-breasted Flycatchers. In the Vienna Woods the species showed a preference for old stands (>100 years), and avoided younger ones (<44 years). To benefit Red-

breasted Flycatchers, stands should be managed less intensively and the cutting cycle should be lengthened. A large leaf surface used for foraging and a high availability of nesting places affect the territorial choice of Red-breasted Flycatchers. Game species seem to have a positive, indirect influence on the population of Red-breasted Flycatchers, as they inhibit growth of the bush layer and ground cover and stimulate the growth of side shoots in hornbeams. In future more attention should be paid to the influence of mammalian game, as their activity can have major implications for nature conservation efforts for species like Red-breasted Flycatchers.

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